

# NUMERICALS

## MPI

Q1) An MPI program runs with  $-np\ 6$ . How many processes execute the program & how many times will a `printf()` inside `main()` run?  
 $np = \text{no. of processes}$   
`mpirun -np 6 ./program`

- 6 processes are created
- If `printf()` is written once inside `main()` & not inside any condition, it will execute once per process
- So `printf()` will run 6 times

Q2) For  $\text{value} = \text{rank} + 1$ , find the result of `MPI_Reduce(sum)` for 4 processes  
 no. of processes = 4

Ranks = 0, 1, 2, 3

Value = rank + 1

Values per process:

| Rank | Value = rank + 1 |
|------|------------------|
| 0    | 0 + 1 = 1        |
| 1    | 1 + 1 = 2        |
| 2    | 2 + 1 = 3        |
| 3    | 3 + 1 = 4        |

`MPI_Reduce` with sum =  $1 + 2 + 3 + 4 = 10$

Code:-

```
MPI_Reduce(&value, &sum, 1, MPI_INT,
           MPI_SUM, 0, MPI_COMM_WORLD);
```

Q3) Root process scatters array [5, 10, 15, 20, 25] among 5 processes. What does each process receive?

no. of processes = 5. Each process will receive one array element

Rank

Received value

0  
1  
2  
3  
4

5  
10  
15  
20  
25

Code:-  

```
int data[5] = {5, 10, 15, 20, 25};
MPI_Scatter(data, 1, MPI_INT, &recv, 1, MPI_INT,
            0, MPI_COMM_WORLD);
```

Q4) Each process sends  $\text{rank} \times 2$  to root using `MPI_Gather`. Find the gathered array for 4 processes  
 no. of processes = 4  
 each process sends  $\text{rank} \times 2$

Rank

0  
1  
2  
3

value = rank \* 2

0  
2  
4  
6

Code:-

`MPI_Collect(8value, 1, MPI_INT, 1, MPI_INT, 0, MPI_COMM_WORLD);`

Q5) Process 0 sends 50 to process 1, process 1 multiplies 4. What is the final output?

→ Process 0 sends 50 to process 1

→ Process 1 multiplies it by 4

$50 \times 4 = 200$   
final output = 200

Q6) How many messages are exchanged in MPI\_Bcast for 8 processes

Messages = processes - 1

= 8 - 1

= 7

Code: `MPI_Bcast(8data, 1, MPI_INT, 0, MPI_COMM_WORLD);`

Code:- `int rank, value;`

`MPI_Init(&argc, &argv);`

`MPI_Comm_Rank(MPI_COMM_WORLD, &rank);`

`if (rank == 0) {`

`value = 50;`

`MPI_Send(&value, 1, MPI_INT, 1, 0, MPI_COMM_WORLD);`

`}`

`else if (rank == 1) {`

`MPI_Recv(&value, 1, MPI_INT, 0, 0, MPI_COMM_WORLD, MPI_STATUS_IGNORE);`

`value *= 4;`

`printf("Result = ", value);`

`}`

`MPI_Finalize();`



# FAULT TOLERANCE

Q1) MTTF = 1200 hrs MTTR = 30 hrs, Calculate availability

$$\text{availability} = \frac{\text{MTTF}}{\text{MTTF} + \text{MTTR}}$$

$$= \frac{1200}{1200 + 30}$$

$$\text{So availability} = 0.9756 \approx 97.5\%$$

Q2) If availability is 0.98 & MTTR = 20 hrs, calculate MTTF

$$\text{availability} = \frac{\text{MTTF}}{\text{MTTF} + \text{MTTR}}$$

$$0.98 = \frac{\text{MTTF}}{\text{MTTF} + 20}$$

$$0.98 (\text{MTTF} + 20) = \text{MTTF}$$

$$0.98 \text{MTTF} + 19.6 = \text{MTTF}$$

$$19.6 = \text{MTTF} - 0.98 \text{MTTF}$$

$$19.6 = \text{MTTF} (1 - 0.98)$$

$$19.6 = \text{MTTF} (0.02)$$

$$\frac{19.6}{0.02} = \text{MTTF}$$

$$\text{MTTF} = 980$$

Q3) A system fails every 600 hrs on average. What is its MTTF?

MTTF = time b/w two failures so

$$\text{MTTF} = 600 \text{ hrs}$$

Q4) If MTTR increases from 10 hrs to 40 hrs, how does availability change (qualitative + numeric question)

MTTR increase from 10 to 40 hrs

→ availability decreases

→ assume MTTF = 1000 hrs

$$\text{availability} = \frac{1000}{1000 + 10} = 0.99 \approx 99\%$$

if "

$$\text{availability} = \frac{1000}{1000 + 40} = 0.962 \approx 96\%$$

availability decreases from 99% to 96%

# DISTRIBUTED SYSTEM & COMMUNICATION

Q1) A system has 10 nodes. Each node sends a heartbeat every 5 seconds. How many heartbeat messages are sent per minute?

→ no. of nodes = 10

→ each node sends 1 heartbeat every 5 seconds

heartbeats per minute =  $\frac{60}{5} = 12$  heartbeats per node

Total heartbeats =  $10 \times 12$   
 $= 120$

Q2) If network latency is 40ms & acknowledgement time is 20ms. calculate round-trip time

network latency = 40ms

acknowledgment time = 20ms

RTT =  $40 + 20$

$= 60\text{ms}$

Q3) A message is retransmitted occur in 3 seconds timeout. How many retransmissions occur in 30 seconds?

timeout = 3 seconds

total time = 30 s

$\frac{30}{3} = 10$  retransmissions

Q4) Horizontal fragmentation splits a table of 1000 records into 5 nodes equally. How many records per node?

total records = 1000

no. of nodes = 5

records per node =  $\frac{1000}{5}$

$= 200$

Q5) Vertical fragmentation splits 10 attributes into 2 fragment equally. How many attributes per fragment?

total attributes = 10

no. of fragments = 2

attributes per fragment = 5



Q5) two independent systems have reliability 0.9 each. What is combined reliability if both must work?  
Reliability of each system = 0.9

$$R = R_1 \times R_2$$

$$= 0.9 \times 0.9$$

$$R = 0.81 \approx 81\%$$

Q6) What is the availability of a system that is down 5 hrs per month? 24x30  
total time = 30 days  
= 720 hrs

$$\text{Availability} = \frac{\text{total time} - \text{down time}}{\text{total time}}$$

$$= \frac{720 - 5}{720}$$

$$= 0.993 \approx 99.3\%$$

## REPLICATION DMR & TMR

Q1) If reliability of one module is 0.7, calculate reliability of DMR

$$\text{Reliability} = 0.7$$

DMR = Dual Modular Redundancy

$$R_{DMR} = R \times R$$

$$= 0.7 \times 0.7$$

$$= 0.49 \approx 49\%$$

Q2) If reliability of one module is 0.8, calculate reliability of TMR

$$\text{using } R_{TMR} = 3R^2 - 2R^3$$

$$R_{TMR} = 3(0.8)^2 - 2(0.8)^3$$

$$= 0.896 \approx 89.6\%$$

Q3) How many failures can TMR tolerate?

→ TMR has 3 modules

→ majority voting requires 2 modules to be working

→ TMR can tolerate one failure

Q4) How many modules are required required to tolerate 2 failures using majority voting?

$$\rightarrow \text{modules required} = 2f + 1 \quad \text{where } f \text{ is no. of failures}$$

$$= 2(2) + 1$$

$$= 5$$



Q5) If one replica fails every 200hrs <sup>how</sup> often does a replicated system of 3 nodes fail?  
 failure time = 200hrs  
 for 3 nodes =  $3 \times 200\text{hrs}$   
 $= 600\text{hrs}$

## CONSENSUS & BYZANTINE FAULTS

Q1) What is minimum no. of processes req. to tolerate 1 Byzantine fault?  
 where  $f$  is no. of Byzantine faults

$$n = 3f + 1$$

$$= 3(1) + 1$$

$$n = 4 \text{ processes}$$

Q2) In a system of 10 processes how many Byzantine faults can be tolerated

$$n = 3f + 1$$

$$n - 1 = 3f$$

$$f = \frac{n-1}{3}$$

$$= \frac{10-1}{3}$$

$$= \frac{9}{3}$$

$$f = 3 \text{ Byzantine faults}$$

Q3) For  $f=3$  Byzantine faults, calculate min req. processes

$$n = 3f + 1$$

$$= 3(3) + 1$$

$$= 9 + 1$$

Q4) If 7 processes vote  $\{1, 1, 1, 0, 0, 0, 0\}$  what value is chosen by majority?

$$\text{no. of 1s} = 3$$

$$\text{no. of 0s} = 4$$

So majority value = 0

Q5) How many rounds of messages exchange are req. in synchronous Byzantine agreement.

General case =  $f+1$  rounds

$$\text{If } f=1 \rightarrow 1+1 = 2 \text{ rounds}$$

$$\text{If } f=2 \rightarrow 2+1 = 3 \text{ rounds}$$

So it will be  $f+1$  rounds



# FAILURE DETECTION

Q1) Heartbeat interval = 2sec, timeout = 6sec  
How many missed heartbeats indicate failure?  
$$\frac{\text{timeout}}{\text{heartbeat interval}} = \frac{6}{2} = 3 \text{ missed heartbeat failures}$$

Q2) If failure detector falsely suspects 2 times 1 minute, is it strongly accurate?

- Strongly accurate req. no false suspicions
- As false suspicions occur, strong accuracy is violated
- No, it's not strongly accurate

Q3) If every crashed process is eventually detected by at least one correct process. Classify completeness

- Every crash failure is eventually detected by at least one correct process.
- The completeness is eventually suspected by some correct process. *weaker completeness*

Q4) How long does failure detection take if timeout is 10sec?

timeout = 10 sec

Detection occurs after timeout expires = 10 sec

Q5) How many messages are req. for heartbeat-based detection in 5 nodes?

- Each node sends to 4 others
- total messages =  $n(n-1)$   
 $= 5(5-1)$   
 $= 20 \text{ messages}$

## GPU & CUDA

Q1) Kernel launch: `kernel<<<8, 256>>>() ;` calculate no. of threads

no. blocks = 8

threads per block = 256

total no. of threads =  $8 \times 256$   
 $= 2048$

Q2) Each thread processes one element. How many elements are processed?  
each thread process 1 element  
Total threads = 2048



Q3) Array size = 2048 floats, float = 4 bytes. How many GPU memory is req?  
array size = 2048  
size of float = 4 bytes

Q4) Global memory latency = 500 cycles, shared memory latency = 25 cycles.  
Calculate speedup

$$\text{Speedup} = \frac{\text{Global memory latency}}{\text{Shared memory latency}}$$

$$= \frac{500}{25}$$

$$= 20\times$$

Q5) CUDA grid has 10 blocks, each block has 128 threads. How many warps are created (assume warp = 32 threads)?

$$\text{Threads per block} = \frac{128}{32}$$

$$= 4 \text{ warps per block}$$

$$\text{Total warps} = 10 \times 4$$

$$= 40 \text{ warps}$$

Q6) If GPU kernel execution takes 2ms & data transfer takes 8ms, what is total execution time?

$$\text{Kernel execution} = 2\text{ms}$$

$$\text{Data transfer} = 8\text{ms}$$

$$\text{Total time} = 2 + 8$$

$$= 10\text{ms}$$

## VIRTUALIZATION & CONTAINERIZATION

Q1) VM startup time = 90s, container startup time = 6sec  
How many times faster is container startup?

$$\text{Speedup} = \frac{\text{VM time}}{\text{Container time}} = \frac{90}{6} = 15\times$$

Q2) A server runs 20 VMs each uses 2GB RAM. How much RAM is required?  
Total RAM =  $20 \times 2$   
 $= 40$

Q3) If container overhead is 5% & VM overhead is 30%. calculate efficiency difference

$$\text{Container overhead} = 5\% \rightarrow \text{efficiency} = 95\%$$

$$\text{VM overhead} = 30\% \rightarrow \text{efficiency} = 70\%$$

$$\text{Difference} = 95\% - 70\% \\ = 25\%$$



Q4) If virtualization reduces hardware usage by 40%. how much is saved from 100 units?

$$\text{Saved} = 100 \times 40\%$$

$$= 100 \times 0.4$$

$$= 40 \text{ units}$$

Q5) A cloud server runs 50 containers, each using 200MB RAM. Calculate total RAM usage.

$$= 50 \times 200$$

$$= 10,000 \text{ MB}$$

$$= 10 \text{ GB}$$

## CHECKPOINT & ROLLBACK

Q1) Checkpoint interval = 15 min, checkpoint overhead = 3 min. Calculate overhead percentage.

$$\text{Overhead\%} = \frac{\text{checkpoint time}}{\text{interval}} \times 100$$

$$= \frac{3}{15} \times 100$$

$$= 20\% \text{ overhead}$$

Q2) System fails after 60 min. How much computation is lost if last checkpoint was at 45 min?

$$\text{Lost computation} = 60 - 45$$

$$= 15 \text{ min}$$

Q3) How many checkpoints occur in 2 hrs with interval 20 min?

$$\text{Interval} = 20 \text{ min}$$

$$\text{Total time} = 2 \text{ hrs}$$

$$= 120 \text{ min}$$

$$\text{No. of checkpoints} = \frac{120}{20}$$

$$= 6 \text{ checkpoints}$$

Q4) If checkpoint size is 200MB & disk speed is 100MB/s. how long does checkpointing take?

$$\text{Time} = \frac{\text{size}}{\text{speed}}$$

$$= \frac{200}{100}$$

$$= 2 \text{ s}$$

Q. If rollback recovery takes 10s, how does it affect MTTR?

Rollback recovery = 10s

MTTR includes these 10s plus any other repair/detection time.

New MTTR =  $X + 10s$



# NUMERICALS

## 1. VIRTUALIZATION

### Q1) RAM allocation

i) Host RAM = 64GB

VMs require 16, 20, 8, 12GB

Calculate total & say if they fit

$$= 16 + 20 + 8 + 12$$

$$= 56GB$$

$56 < 64$  easily fits

ii) Host RAM = 32GB

4 VMs require 8, 12, 6, 8GB

Will all RAMs run simultaneously?

$$= 8 + 12 + 6 + 8$$

$$= 34GB$$

$34 > 32$  can't allocate all VM's simultaneously

### Q2) CPU allocation

i) Host has 8 vCPUs

VMs require 1.5, 2.0, 0.75, 1.25, 2.0 CPUs

Find total CPU demand & say if overcommit occurs.

$$= 1.5 + 2.0 + 0.75 + 1.25 + 2.0$$

$$= 7.5 \text{ CPUs}$$

$7.5 < 8$  No overcommit

ii) A hypervisor schedules 4 VMs on 8 CPUs

Each VM require 0.5, 0.75, 0.25, 0.6 CPU

What is total CPU utilization?

$$= 0.5 + 0.75 + 0.25 + 0.6$$

$$= 2.1$$

$2.1 > 2$  There's a slight overcommit of 0.1.

### Q3) Virtualization overhead on execution time

i) Program runs 100ms on bare metal.

Virtualization overhead = 12%. Find VM execution time.

$$\text{Multiplier} = 1 + 0.12$$

$$= 1.12$$

$$100 \times 1.12 = 112 \text{ms}$$

ii) A VM runs an application that takes 50ms on physical hardware. Virtualization overhead is 15%. Find execution time.

$$\text{Multiplier} = 1 + 0.15$$

$$= 1.15$$

$$\text{Execution time} = 50 \times 1.15$$

$$= 57.5 \text{ms}$$

### Q4) Network throughput under hypervisor overhead

Physical network = 1000 Mbps. Type-1 overhead = 2%.

Type-2 overhead = 8%.

Compute effective throughput.

$$\text{Type-1: } 1000 \times (1 - 0.02) = 1000 \times 0.98 = 980 \text{Mbps}$$

$$\text{Type-2: } 1000 \times (1 - 0.08) = 1000 \times 0.92 = 920 \text{Mbps}$$



## 2. CONTAINERIZATION

### Q1) Memory Packing

Host RAM = 32GB

Containers with memory 0.5, 0.75, 1, 0.25, 0.5, 0.5, 0.75  
Compute total memory used & how many smallest (0.25GB) containers could fit max

$$\text{Total} = 0.5 + 0.75 + 1 + 0.25 + 0.5 + 0.5 + 0.75 \\ = 4.25 \text{ GB}$$

Smallest container = 0.25

$$\text{Max identical small container} = 32 / 0.25 \\ = 128 \text{ containers}$$

### Q2) Startup time Comparison

VM startup = 30s

Container setup = 2s

How many containers start in same time as 1 VM?

$$= 30s / 2s$$

$$= 15 \text{ containers}$$

## 3. HYPERVISOR TYPE-I, TYPE-II

### Q1) CPU utilization with hypervisor

Host: 16 cores

Type-I overhead = 3%

CPU reserved for hypervisor

Type-II overhead = 7% for host OS + hypervisor

Available cores effectively

$$\text{Type-I available} = 16 \times (1 - 0.03) \\ = 16 \times 0.97 \\ = 15.52 \text{ cores}$$

$$\text{Type-II available} = 16 \times (1 - 0.07) \\ = 16 \times 0.93 \\ = 14.88 \text{ cores}$$

## 5. DISTRIBUTIVE SYSTEMS

### Q1) Amdahl's Law (speedup)

Total job time = 200s Fraction sequential = 0.2  
(0.2). Using 5 nodes, compute execution time.

$$\text{Sequential time} = 0.2 \times 200 \\ = 40s$$

$$\text{Parallel portion} = 0.8 \times 200 = 160s \\ \text{on 5 nodes: } 160 / 5 = 32s$$

$$\text{Total} = 40 + 32 = 72s$$

## 4. COMPUTER SYS ARCHITECTURE

### Q1) Privileged Instruction count

Program executes 1,000,000 instructions.

Privileged inst fraction

$$= 1,000,000 \times 0.001$$

$$= 1000 \text{ privileged instructions.}$$

### Q2) ABI binary compatibility

Binary has 3 sections

text = 2.1MB

data = 0.4MB

bss = 0.5MB

Compute total binary size

$$= 2.1 + 0.4 + 0.5$$

$$= 3 \text{ MB}$$

### Q2) Communication Time

Send 800MB over link bandwidth

200MB/s with latency 0.1s Time = latency + size/bw.

$$= 0.1 + (800 / 200)$$

$$= 0.1 + 4$$

$$= 4.1s$$



### Q3) Replication update time

There are 4 replicas, updating each replica takes 0.15s sequentially. Total update time?

$$= 4 \times 0.15$$

$$= 0.6s$$

### Q5) Vertical partition example

Table columns size per row A=200, B=20, C=5 KB. If vertical split stores A, B on server1 & C on server2 & you need full row of 1,000,000 rows compute data size.

$$\text{Server1 size} = (200+20) \times 1,000,000$$

$$= 220 \times 1,000,000$$

$$= 220,000,000 \text{ units}$$

$$\text{Server2 size} = 5 \times 1,000,000$$

$$= 5,000,000 \text{ units.}$$

## 6. MPI

### Q1) All to all message count

N=6 processes performing all to all (each sends to every other) find total messages.

$$= N \times (N-1)$$

$$= 6 \times (6-1)$$

$$= 6 \times 5$$

$$= 30 \text{ messages}$$

### Q3) Scatter (Sequential root sends)

Root sends 8 chunks, each 2MB, bandwidth=100MB/s, latency=0.01s, seq sends. Time =  $N \times (\text{latency} + \text{chunks}/\text{bw})$

$$\text{Time} = 8 \times (0.01 + 2/100)$$

$$= 8 \times (0.01 + 0.02)$$

$$= 8 \times (0.03)$$

$$\text{Time} = 0.24s$$

### Q4) Data Placement

Database size 12TB. Horizontal sharding into 6 shards. Find size per shard?

$$= 12TB/6$$

$$= 2TB \text{ per shard}$$

### Q2) Point-to-Point time

Message size=500kb, bandwidth=1000KB/s,

latency=0.005s. Time = latency + size/bw

$$\text{Time} = 0.005 + 500/1000$$

$$= 0.005 + 0.5$$

$$\text{Time} = 0.505s$$

### Q4) Reduce (Tree cost model)

P=16 processes, message size=4MB,

bandwidth=100MB/s, latency=0.01s.

for a binary tree Reduce cost  $\approx \log_2(P) \times (\text{latency} + \text{msg}/\text{bw})$

$$\text{Reduce cost} = \log_2(16) \times (0.01 + 4/100)$$

$$= 4 \times (0.01 + 0.04)$$

$$= 4 \times 0.05$$

$$= 0.20s$$



## Q5) Parallel speedup with MPI

Sequential time = 100s. Work split equally among 5 processes, comm overhead = 5s

Find total parallel time

$$\begin{aligned}\text{Total parallel time} &= (100/5) + 5 \\ &= 20 + 5 \\ &= 25\text{s}\end{aligned}$$

$$\begin{aligned}\text{Speedup} &= 100 / 25 \\ &= 4\times\end{aligned}$$

## 7. MPI FUNCTIONS

Q)  $P=8$  processes. Root = rank 0. You have an array of 80 integers (10 per process logically). Network latency = 0.005s, bandwidth = 500 ints/s (assume 1 int = 4 bytes but treat bandwidth in ints/s for simplicity).

### 1. MPI\_COMM\_WORLD

membership count:  $P=8$  so operations that loop over MPI\_COMM\_WORLD iterate 8 ranks.

2. MPI\_Scatter - root sends 10 ints to each process (including itself)

$$\begin{aligned}\text{Per chunk time} &\approx \text{latency} + \text{chunk/bw} \\ &= 0.005 + 10/500 \\ &= 0.005 + 0.02 \\ &= 0.025\text{s}\end{aligned}$$

$$\begin{aligned}\text{If root sends sequential sends, total} &= 8 \times 0.025 \\ &= 0.20\text{s}\end{aligned}$$

3. MPI\_Bcast - broadcasting 80 ints from root to all  $P$ .

$$\begin{aligned}\text{Using binomial tree cost} &\approx \log_2(P) \times (\text{latency} + \text{msg/bw}) \\ &= \log_2(8) \times (0.005 + 80/500) \\ &= 3 \times (0.005 + 0.16) \\ &= 3 \times (0.165)\end{aligned}$$

$$\text{Time} = 0.495\text{s}$$

4.) MPI\_Gather - each process sends its 10 ints to root

$$\begin{aligned}\text{Sequential gather cost} &= 8 \times (0.005 + 10/500) \\ &= 8 \times (0.005 + 0.02) \\ &= 8 \times 0.025 \\ &= 0.2\text{s}\end{aligned}$$